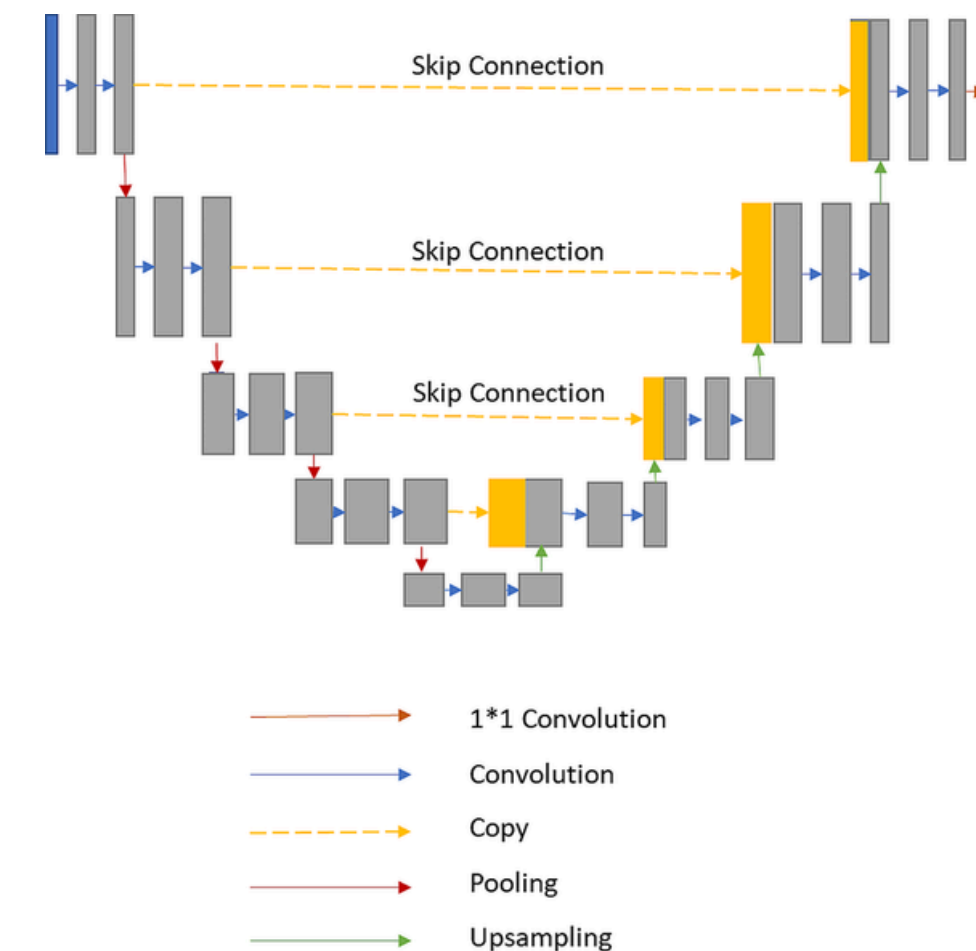


Task: Radar Sematic Segmentation

- The Challenge: Identify and classify objects (Suitcase, Chair, Human, Wall) from raw 6-channel radar data.
- Data Complexity: Radar signals are significantly noisier than standard RGB images, featuring "ghost" reflections and motion blur that obscure targets.
- High-Stakes Scoring: The IOAI contest uses a weighted metric where background pixels are worth 1 point, but object pixels are worth 50 points, making missing an object extremely costly.

The Solution – Improved U-Net Architecture

- Encoder-Decoder Design: I utilized a U-Net to capture both high-level context and fine-grained spatial details.
- Skip Connections: These "shortcuts" allow the model to preserve exact pixel locations of small objects, which is critical for achieving high accuracy in the 50-point categories.
- Strategic Regularization: We integrated Batch Normalization and Dropout in the bottleneck to filter out radar noise and prevent overfitting to specific sensor artifacts.
- Class Weighting: The loss function was weighted at 1:10 to align the model's learning directly with the competition's scoring rewards.



Performance Metrics & Results

Baseline Solution: Validation set score: 0.6205 Testing set score: 0.6217 after 2 minutes training 40 epoch

U-net: Validation set score: 0.9538 Testing set score: 0.9519 after 7.5 minutes of training 40 epoch

ResNet + U-Net: Validation set score: 0.9726 Testing set score: 0.9727 after 3 minutes training 15 epoch

Attention U-Net: Validation set score: 0.9669 Testing set score: 0.9668 after 6 minutes of training 30 epoch

So, as we can see the best model for this task is ResNet + U-Net